Energetic expenditure during walking and load-carriage in humans. Dataset from EVOBREATH 2015

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The Dataset includes data of an experimental research project led by Dr. A. Mateos and approved by the Hospital Universitario de Burgos Ethical Committee (Burgos, Spain) (Ref. CEIC 1480).The aim of the project was to quantify the energetic cost of walking and carrying burdens in a sample of humans, of both sexes, aged from 23 to 50 years old. The sample was exhaustively controlled by means of exclusion criteria aimed to eliminate or minimise the effects of pathologies, medication intake, smoking, food ingestion, or menstrual cycle. Therefore, individuals presenting *diabetes mellitus*, thyroid disease, other neurodegenerative, heart, pulmonary, or skeleto-muscular disease, pregnancy or breastfeeding, active cancer, and medication known to affect metabolism and/or endocrine function, were excluded from the sample. Participants were required to fast overnight prior to having their metabolic measurements, taken in the morning.

The experiment was conducted at the at the Bioenergy Laboratory of the National Research Centre on Human Evolution (CENIEH, Burgos, Spain) under standard conditions of temperature (21.5ºC-23ºC) and humidity (29%-31%). Several protocols were applied.

- ✓ The Anthropometric Protocol was performed following normalised standards. Height was measured nearest to 0.1 cm with a Harpenden stadiometer (Holtain Limited). For body segments (bi-iliac breadth and femur length), an anthropometer (Harpenden, Holtain Limited) was used. Body mass (Weight) was obtained with a digital scale to the nearest 0.1 kg.

    - Femur length (FL) was measured according to the trochanteric length (the distance between the greater trochanter and the femoral lateral condyle), described by Lavelle (1974).

    - Bi-iliac breadth (BIL) was measured as the distance between the most lateral points of the iliac tubers.

- ✓ The Body Composition Protocol was applied, following normalised standards, with an analyzer vector of body impedance, BIA 101 AKERN©, and the software BodyGram Pro© (v2010). The proportion of fat mass (FM) and fat-free mass (FFM) was obtained.

- ✓ The indirect calorimetry protocol was carried out with an ergospirometer Master Screen CPX JAEGER™ and the LabManager© IntelliSupport 5.2 software application. Resting Metabolic Rate was measured during 30 minutes, with each participant layed back on a stretcher. Volunteers performed 4 different locomotion tests on an hp COSMOS© treadmill with no inclination and a constant speed of 4 km/h for 11 minutes. Walking weight trials: (WW00) walking

unladen; (WW05) walking carrying a backload of 5 kg; (WW10) walking carrying a backload of 10 kg; (WW15) walking carrying a backload of 15 kg.

The dataset contains the following variables:

Variable	Description	Unit
ID	Identification code of the subject	
Sex	M (Male) or F (Female)	
Age	Age of the subject	years
Height	Stature	cm
Weight	Body mass	kg
FL	Femur length	cm
BIL	Bi-iliac breadth	cm
FM	Fat Mass	kg
FFM	Fat-Free Mass	kg
RMR	Resting Metabolic Rate	kcal/day
EE_WW00	Energy expenditure Walking unladen	kcal/day
EE_WW05	Energy expenditure Walking carrying 5kg	kcal/day
EE_WW10	Energy expenditure Walking carrying 10kg	kcal/day
EE_WW15	Energy expenditure Walking carrying 15kg	kcal/day

References

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